

Maximize $x_1 + x_2$ subject to

$$\begin{cases} -x_1 + x_2 \leq -1 \\ -x_1 + 3x_2 \leq 3 \end{cases} \quad x_1 \geq 0 \quad x_2 \geq 0$$

Will a solution exist if the objective function is changed to $z = -x_1 - x_2$?

Solve wherever possible.

[This question paper contains 8 printed pages.]

04.06.24 (M)

Your Roll No.

Sr. No. of Question Paper : 1608

H

Unique Paper Code : 12271202

Name of the Paper : Mathematical Methods for Economics-II

Name of the Course : B.A. (Hons.) Economics

Semester : II

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. There are 4 questions in all.
3. All questions are compulsory.
4. All parts of a question must be answered together.
5. Use of simple calculator is allowed.

1. Answer any **four** of the following : (4×5=20)

(a) The marginal cost (in lakhs of rupees) of producing a motor car is given by $6 + 4x^2 + 1.5e^x$, where x is the quantity produced. Determine the total cost producing 5 motor cars if the fixed cost is Rs. 7 Lakhs. It is given that $e^x = 0.006$.

(b) (i) Suppose $Y = Y(t)$ is national product, $C(t)$ is consumption at time t , and $\bar{I}$ is Investment, which is constant. Suppose $\dot{Y} = \alpha(C + \bar{I} - Y)$ and $C = aY + b$, where a , b and α are positive constants with $\alpha < 1$. Derive a differential equation for Y .

(ii) Find its solution when $Y(0) = Y_0$ is given. What happens to $Y(t)$ as $t \rightarrow \infty$?

(c) Let Y_t denote national income, I_t total investment, and S_t total saving – all in period t . Suppose savings are proportional to national income, and that investment is proportional to the change in income from period t to $t+1$. Then, for $t = 0, 1, 2, \dots$,

(e) Find the maximum and minimum values of $f(x, y) = x + y$ subject to $x^2 + y^2 = 4$ using Lagrange multiplier method. Verify the second order conditions.

4. Answer any **one** of the following : (1×7=7)

(a) The Eco club Econ manufactures eco-friendly inexpensive notebooks and diaries. Each notebook takes 4 Kgs of paper pulp and 2 litres of water solution in its preparation while each diary requires 3 Kgs of paper pulp and 1 litre of water solution. The club has 240 Kgs of paper pulp and 100 litres of water solution available with it. Each notebook sold gives a profit of Rs. 7 and each diary gives a profit of Rs. 5. Using linear programming, find the optimal combination of notebooks and diaries to be produced in order to maximize the profit.

(b) Is there a solution to the following linear programming problem?

Point	f'_x	f'_y	f''_{xx}	f''_{xy}	f''_{yy}
A	1	2	3	4	5
B	0	0	0	0	0
C	0	0	1	2	3
D	0	0	0	1	0
E	0	0	0	-1	0

- (b) Find the extreme points of the function $f(x,y) = 9x + 8y - 6(x+y)^2$ subject to $0 \leq x \leq 5$, $0 \leq y \leq 3$ and $x - 2y \geq -2$.

- (c) Find all extreme points and/or saddle points of the function :

$$3x^2y + y^3 - 3x^2 - 3y^2 + 2$$

- (d) Define quasi-concave and quasi-convex functions. Prove geometrically or otherwise that $y = x^2$: $x \geq 0$ is both quasi-concave and quasi-convex.

$$(i) S_t = \beta Y_t$$

$$(ii) I_{t+1} = \alpha(Y_{t+1} - Y_t)$$

$$(iii) S_t = I_t \quad \alpha, \beta > 0 \quad \beta < \alpha$$

Deduce a difference equation determining the path of Y_t , given Y_0 and solve it.

- (d) Find the general form of a function f whose second derivative $f''(x)$ is equal to x^2 . If we require in addition that $f(0) = -1$, what is $f(x)$?

- (e) Find the area of the region bounded by $y = x^2$ and $y = x$.

2. Answer any **four** of the following : (4×5=20)

- (a) Show that the production function $Y = Ax^\alpha y^\beta$ where $A, \alpha, \beta > 0$ has elasticity of substitution 1 everywhere.

- (b) A firm sells 2 brands X and Y of a soap. The outputs and prices are denoted by x, y, p and q respectively. The demand functions of the two

brands are : $x = 100 - 2p + 5q$ and $y = 80 + 4p - 3q$. Suppose brand X sells for rupees 15 per unit and Y for 12 per unit. Estimate the approximate change in revenue if the prices are increased by 1 per unit for X and 1.5 per unit for Y and compare it with the actual change in revenue.

(c) Draw the level curves for the function

$$f(x, y) = \frac{2x - 2y}{x^2 + y^2 + 1} \text{ for levels } k = 0, 1.$$

[For PWD candidates in lieu of part c]

Find the equation of the tangent plane at a point $(1, 1, 5)$ to the function $x^2 + 2xy + 2y^2$

(d) Examine the definiteness of the following quadratic forms :

(i) $f(x, y) = x^2 + 2xy + y^2$

(ii) $-x^2 + y^2$ subject to $2x - 3y = 5$

(e) Find and sketch the domain of the following functions :

(i) $f(x, y) = \sqrt{x^2 - y}$

(ii) $f(x, y) = \sqrt{y+4} - \sqrt{x+1}$

[For PWD candidates in lieu of part e]

Find the domain of the following functions :

(i) $f(x, y) = \sqrt{x^2 - y}$

(ii) $f(x, y) = \sqrt{y+4} - \sqrt{x+1}$

(iii) $f(x, y) = \ln(y^2 - x)$

3. Answer any **four** of the following : $(4 \times 7 = 28)$

(a) Based on the table, classify each of the following points as a local maximum, local minimum, saddle point, not a critical point, or not enough information to classify.